6. Vacuum Cleaner Problem

```
class VacuumEnvironment:
    def __init__(self, agent_pos, status_A, status_B):
        self.location = agent_pos  # 'A' or 'B'
        self.status = {'A': status_A, 'B': status_B}  # 'Clean' or 'Dirty'

    def run_agent(self):
        print(f"Initial State -> Location: {self.location}, Status: {self.status}\n")

        # Limit execution loops to avoid infinite loops if an error occurs
        for _ in range(4):
            current_loc = self.location
            current_status = self.status[current_loc]

            print(f"Agent is in Room {current_loc}. Sensor detects room is {current_status}.")

            if current_status == 'Dirty':
                self.status[current_loc] = 'Clean'
                print(f"Action: SUCK dirt in Room {current_loc}. Room updated to Clean.")
            else:
                # If clean, move to the opposite room
                next_loc = 'B' if current_loc == 'A' else 'A'
                print(f"Action: Room {current_loc} is already clean. Move to Room {next_loc}.")
                self.location = next_loc

            # Stop if all locations are verified clean
            if self.status['A'] == 'Clean' and self.status['B'] == 'Clean':
                print(f"\nFinal State Achieved -> Status: {self.status}. All rooms clean!")
                break

if __name__ == "__main__":
```

Start Agent in Room A, where A is Dirty and B is Dirty

env = VacuumEnvironment(agent_pos='A', status_A='Dirty', status_B='Dirty')

env.run_agent()

main.py Visualize Run

```
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    def __init__(self, agent_pos, status_A, status_B):
        self.location = agent_pos  # 'A' or 'B'
        self.status = {'A': status_A, 'B': status_B}  # 'Clean' or 'Dirty'

    def run_agent(self):
        print(f"Initial State -> Location: {self.location}, Status: {self.status}\n")

        # Limit execution loops to avoid infinite loops if an error occurs
        for _ in range(4):
            current_loc = self.location
            current_status = self.status[current_loc]

            print(f"Agent is in Room {current_loc}. Sensor detects room is {current_status}.")

            if current_status == 'Dirty':
                self.status[current_loc] = 'Clean'
                print(f"Action: SUCK dirt in Room
```

Output

```
Initial State -> Location: A, Status: {'A': 'Dirty', 'B
'Dirty'}

Agent is in Room A. Sensor detects room is Dirty.
Action: SUCK dirt in Room A. Room updated to Clean.
Agent is in Room A. Sensor detects room is Clean.
Action: Room A is already clean. Move to Room B.
Agent is in Room B. Sensor detects room is Dirty.
Action: SUCK dirt in Room B. Room updated to Clean.

Final State Achieved -> Status: {'A': 'Clean', 'B':
'Clean'}. All rooms clean!
```